Tao Wang

Moscow, Idaho | twang1217.github.io

EDUCATION

Doctor of Philosophy in Computer Science January 2024 – Present
University of Idaho, Moscow, Idaho

Bachelor of Science in Computer Science August 2022 – May 2023
University of Idaho, Moscow, Idaho

RESEARCH EXPERIENCE

Graduate Research Assistant January 2024 – August 2026
Department of Computer Science, University of Idaho, Moscow, Idaho

Machine Learning for Nuclear Materials Properties (with Idaho National Laboratory)

- Proposed a temperature-dependent, geometry-corrected stress descriptor for machine learning prediction of creep rupture life; first-author manuscript under review at *Journal of Nuclear Materials*.
- Developed a domain-adaptation machine learning approach that correlates full-sized Charpy impact properties with sub-sized specimen tests; first-author manuscript under review.
- Contributed to a public, machine-learning-ready creep-property dataset released with open-source code; published in *Scientific Data*.

LLM Framework for Nuclear Power Plant Event Classification (with Idaho National Laboratory)

- Co-developed a two-stage pipeline pairing knowledge-informed prescreening (44 domain-specific text patterns) with a BERT-based LLM to classify shutdown initiating events in U.S. nuclear power plant reports for probabilistic risk assessment.
- Filtered out more than 97% of irrelevant records and achieved 93.4% average accuracy across four event types; published in the *Journal of Risk and Reliability*.
- Co-developed a web-based tool implementing the framework and deployed it to Idaho National Laboratory.

PROJECTS

Mars Reentry Vehicle Senior Capstone (with NASA Ames Research Center) August 2022 – May 2023

- Collaborated on a seven-person interdisciplinary team to design, build, and flight-test a tube-deployed reentry vehicle prototype for a potential Mars weather station; won Best Booth Presentation at EXPO 2023.
- Validated the prototype through a high-altitude balloon drop (6,688 m apogee, 12.7 m/s terminal velocity, 0.90 drag coefficient) using an ESP32 flight computer running FreeRTOS.

PUBLICATIONS

- **Wang, T.**, et al. (under review). A temperature-dependent geometry-corrected stress descriptor for machine learning-based prediction of creep rupture life. *Journal of Nuclear Materials*.
- **Wang, T.**, et al. (under review). Domain-adaptation machine learning approach for correlation of full-sized Charpy properties from sub-sized tests.
- Xian, M., **Wang, T.**, et al. (2025). A knowledge-informed large language model framework for U.S. nuclear power plant shutdown initiating event classification for probabilistic risk assessment. *Journal of Risk and Reliability*, 239(6), 1257–1264.
- Lee, I., …, **Wang, T.**, et al. (2026). Compilation of creep property data for nuclear structural materials. *Scientific Data*, 13, 1262.

SKILLS

Machine Learning: Large language models, NLP, text classification, BERT fine-tuning, domain adaptation, feature engineering, imbalanced and noisy-label learning

Domain Knowledge: Nuclear power plant safety, probabilistic risk assessment, nuclear structural materials (Charpy impact and creep properties), scientific data curation